Fiscal Year:	FY 2022	**Task Last Updated:**	FY 03/01/2022
PI Name:	Gonzalez, Adam Ph.D.		
Project Title:	Asynchronous Techniques for the Delivery of Empirically Supported Psychotherapies		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Stony Brook	**State:**	NY
Zip Code:	11794-3365	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-Crew Health (FLAGSHIP & NSBRI)
Start Date:	07/01/2015	**End Date:**	06/30/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date is now 6/30/2019 per NSSC information (Ed., 6/21/18) NOTE: Element change to Human Factors & Behavioral Performance; previously Behavioral Health & Performance (Ed., 1/18/17)		
Key Personnel Changes/Previous PI:	August 2016: CoInvestigators Charles Boicey and Bireswar Loha have left Stony Brook Unviersity and are no longer working on the project. Their effort has been replaced by post-doctoral fellow (Minjung Shim, PhD).		
COI Name (Institution):	Kaufman, Arie Ph.D. (State University of New York, Stony Brook) Kotov, Roman Ph.D. (State University of New York, Stony Brook) Mahaffey, Brittain Ph.D. (State University of New York, Stony Brook)		
Grant/Contract No.:	NNX15AN64G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Astronauts are exposed to extreme environments and behavioral health risks during space missions. These stressors may place astronauts at elevated risk for behavioral health consequences including delirium, adjustment disorders, asthenia, and mood and anxiety problems, especially during long-duration missions. An expected barrier to delivering treatment to astronauts during long-duration missions is the lag in real-time communication between astronauts and mental health specialists on Earth. Therefore, the objective of this proposal is to identify technologies that will best facilitate the use of psychotherapeutic techniques when real-time communication is not possible. Phase I will involve systematic reviews of the existing literature on (a) psychotherapy techniques currently and historically used on space missions and; (b) existing technologies (e.g., video diaries, virtual reality immersion) with the potential for use in the asynchronous delivery of psychotherapeutic techniques to astronauts. Phase II will use the knowledge gained from Phase I to inform the design of a three-arm randomized clinical trial aimed at evaluating the feasibility, acceptability, and efficacy of the most promising technologies, alone and in combination, for asynchronous delivery of empirically supported psychotherapeutic skills. Products of this study will include best practice guidelines for asynchronous delivery of empirically supported psychotherapeutic techniques during long-duration space missions. These guidelines will be made public via a final report and peer-reviewed manuscript. Further, we will develop a presentation/workshop based on our findings to present to NASA behavioral health experts.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research will help inform best-practice guidelines for asynchronous communication for behavioral heath during long duration space missions.
Task Progress:	In July 2018, the web-based MyCompass program upgraded to MyCompass 2.0 (improvements made to usability but not changes to treatment content). Prior to the MyCompass program upgrade, we had screened a total of 140 individuals from Stony Brook University campus, Stony Brook Hospital, and major science laboratories in the Long Island area. A total of 56 “astronaut-like” participants experiencing elevated psychological distress were randomly assigned to one of the three treatment conditions. We also updated our protocol to allow us to recruit remotely and expand our recruitment search. After MyCompass 2.0 launched in early July 2018, we began to recruit remotely, targeting states in Eastern Standard Time (EST). An additional 89 participants were enrolled remotely from varying Science, Technology, Engineering, and Mathematics (STEM) universities and science laboratories by September 26, 2018. In total, 145 participants were enrolled to the randomized clinical trial (RCT) out of 300 individuals screened. Upon finishing the 7-week treatment, participants completed a post-treatment assessment and an exit interview over the phone. A 1-month follow-up assessment was also administered to participants. All post-treatment and 1-month follow-up assessments have been completed. All survey data was collected via Qualtrics and exported to the Statistical Package for the Social Sciences (SPSS) for analysis. An Excel codebook was created to describe the contents, structure, and layout of the final database. Main study analyses were conducted and best practice guidelines were created and shared with NASA officials.
Bibliography Type:	Description: (Last Updated: 04/11/2022)